AI Usage Report

1. AI Tools Used

ChatGPT (OpenAI) - Used for backend development guidance, debugging, architecture design, and frontend UI support.
Stack Overflow / Documentation - Used for reference and validation.

2. Prompts Used

Examples:
- Design Clean Architecture for .NET Core Employee Management System
- Implement JWT Authentication in .NET Core Web API 10
- Fix SQL Injection risk in repository pattern
- Create responsive login page using Bootstrap
- OWASP Security Considerations

3. Areas Where AI Assisted

Backend: JWT, Authentication, Services, Repository Pattern, Middleware
Frontend: Login UI, Navbar, Validation, Fetch API integration
Architecture: Clean Architecture design, DTO structuring

4. Validation Approach

Manual testing using Swagger and Postman.
Input validation for forms.
Role-based access testing.
Debugging using browser console and logs.
Exception handling verification via middleware.

5. Percentage of AI Generated Code

Backend: 30-40%
Frontend: 30-50%
Debugging: 10%
Architecture: 10%
Overall: ~45% AI assisted development

6. Architecture Overview

API Layer -> Application Layer -> Domain Layer -> Infrastructure Layer -> Persistence Layer
Patterns Used: Clean Architecture, Repository Pattern, Dependency Injection, DTO Pattern

API END POIN OVERVIEWS

Base URLs

https://localhost:{port}/api

Auth Controller

POST /api/auth/login

Description: Authenticates user and returns JWT token.

Request Body:

```
{
  "username": "admin",
  "password": "Admin@123"
}
```

Success Response:

```
{
  "token": "jwt_token",
  "username": "admin",
  "role": "Admin"
}
```

Employee Controller

GET /api/employee

Get all employees (Admin only)

GET /api/employee/{id}

Get employee by ID

POST /api/employee

Create employee

PUT /api/employee/{id}

Update employee

DELETE /api/employee/{id}

Delete employee

GET /api/employee/search

Search employees with filters: keyword, department, designation, isActive

GET /api/employee/dashboard

Get dashboard summary